IRJTAE

Ingenious Research Journal for Technological Advancements in Engineering
(Open Access, Peer-Reviewed, Technological Journal)
Volume:03/Issue:01/March-2026 www.irjtae.com

SIGN LANGUAGE DETECTION DEVICE

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ABSTRACT

This project presents a low-cost wearable Sign Language Detection Device designed to assist individuals with speech or hearing impairments in communicating more effectively. The system uses a specially designed glove embedded with flex sensors to measure finger bending and an MPU6050 sensor to capture hand orientation and motion. These sensors continuously collect data, which is then sent to an Arduino Uno for processing. The Arduino analyzes the sensor values and compares them with predefined gesture patterns to accurately recognize specific hand gestures. Once a gesture is identified, the Arduino transmits the corresponding data wirelessly through an HC-05 Bluetooth module to a mobile application. The mobile app converts the received data into readable text and further uses a text-to-speech (TTS) engine to generate audio output in real time. This enables smooth communication between the user and people who do not understand sign language. The system is portable, efficient, cost-effective, and easy to use, making it suitable for daily life applications. It also provides a strong foundation for future enhancements such as adding more gestures or integrating advanced algorithms.

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Keywords: Sign Language Detection, Wearable Device, Flex Sensors, MPU6050, Arduino Uno, Gesture Recognition, Bluetooth Communication, HC-05 Module, Text-to-Speech (TTS), Assistive Technology, Real-Time Communication, Embedded Systems).

I. INTRODUCTION

Communication is a basic human need, yet individuals with speech and hearing impairments often struggle to interact with people who do not understand sign language. This project addresses that challenge by developing a **wearable sign language detection device** that translates hand gestures into real-time text and speech, making communication easier and more inclusive. The system is built around a smart glove equipped with flex sensors and an MPU6050 sensor. The flex sensors detect the bending of each finger, while the MPU6050 captures hand orientation, tilt, and motion. These sensors continuously generate data that represents different hand gestures. This data is then sent to the Arduino Uno, which acts as the main processing unit. The Arduino analyses the sensor values and compares them with predefined gesture patterns to accurately recognize specific signs. Once a gesture is identified, the Arduino sends the corresponding information wirelessly using an HC-05 Bluetooth module to a mobile application. The mobile app receives this data, displays the recognized gesture as readable text, and converts it into speech using a text-to-speech (TTS) system. This allows the user's hand gestures to be instantly translated into audible communication. The device is designed to be portable, affordable, and easy to use, making it suitable for everyday situations such as classrooms, hospitals, and public places. By reducing the need for human interpreters and enabling real-time interaction, the system improves accessibility, promotes independence, and enhances social inclusion for people with speech and hearing impairments.

. The system is portable, affordable, and helps improve communication, accessibility, and social inclusion.

II. METHODOLOGY

The methodology of this project involves capturing hand gestures using flex sensors and an MPU6050 sensor, which detect finger bending and hand movement. The sensor data is collected and processed by an Arduino Uno to identify predefined gesture patterns. Once a gesture is recognized, the corresponding data is transmitted wirelessly through an HC-05 Bluetooth module to a mobile application. The application then converts the

received data into readable text and generates speech using a text-to-speech (TTS) system. This complete process enables real-time translation of sign language into understandable text and audio output.

Figure 1: Methodology

Data Acquisition and Processing:

The system begins by collecting hand gesture data using flex sensors and an MPU6050 module embedded in the glove. The flex sensors measure the bending of fingers, while the MPU6050 captures hand orientation and motion. This data is sent to the Arduino Uno, where it is processed and analyzed to identify specific gesture patterns based on predefined values.

Gesture Recognition and Output Generation:

Once the gesture is recognized, the Arduino transmits the corresponding data wirelessly through the HC-05 Bluetooth module to a mobile application. The application then converts the received data into readable text and generates speech output using a text-to-speech (TTS) engine, enabling real-time communication between the user and other

III. MODELLING AND ANALYSIS

The system is designed as a step-by-step data processing pipeline that continuously monitors and interprets hand gestures. It starts with the initialization of sensors, where flex sensors measure finger bending and the MPU6050 captures hand orientation, tilt, and motion. These sensors continuously generate raw data while the user performs gestures.

This raw sensor data is then sent to the Arduino Uno, where it undergoes preprocessing. Techniques like filtering and averaging are applied to remove noise, reduce fluctuations, and improve the accuracy of the readings. Clean and stable data is essential for correct gesture recognition. After preprocessing, the system moves to the gesture classification stage, where the processed values are compared with a predefined dataset or threshold-based rules stored in the Arduino. Each gesture has a unique pattern of sensor values, and the system matches the incoming data with these patterns. A decision-making step checks whether a valid gesture is recognized. If the data does not match any known pattern, the system continues collecting new data in a loop, ensuring continuous real-time monitoring. When a gesture is successfully identified, the Arduino converts it into a corresponding digital code or message. This information is then transmitted wirelessly through the HC-05 Bluetooth module to a mobile application. The mobile app receives the data, displays the recognized gesture as text, and uses a text-to-speech (TTS) engine to generate voice output.

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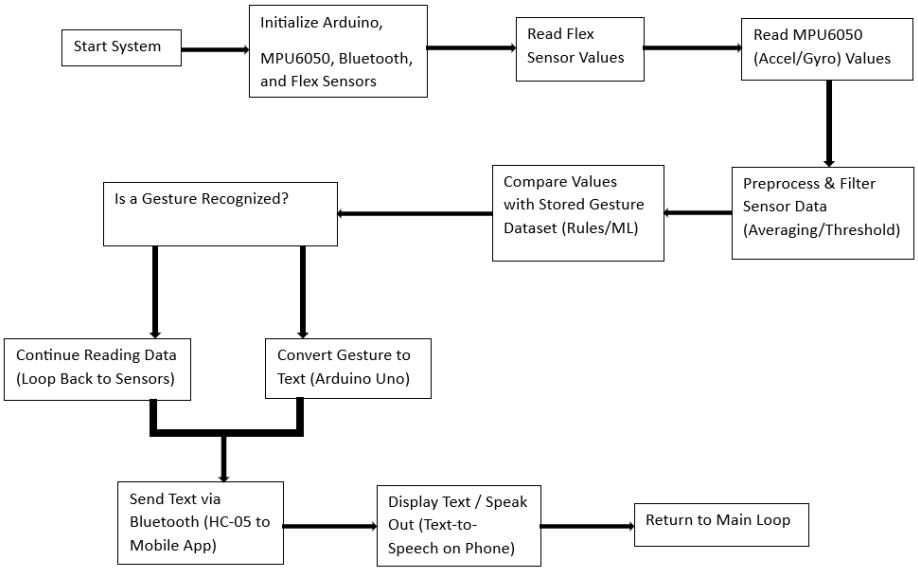

Figure 2: Work-Flow diagram.

IV. RESULTS AND DISCUSSION

The developed Sign Language Detection Device successfully recognizes predefined hand gestures using flex sensors and the MPU6050 module. The system accurately processes the sensor data through the Arduino Uno and displays the corresponding output on the LCD as well as transmits it via Bluetooth to a mobile application. The outputs such as *"I am hungry," "Yes/OK,"* and *"I am thirsty"* confirm that the device can effectively convert gestures into meaningful text. The results demonstrate that the system performs reliably in real time with good accuracy, providing a practical and efficient solution for assisting communication of individuals with speech or hearing impairments.

Table 1. Result: Sign to Messages

SN.	Messages	Thumb	Index Finger	Ring Finger
1	Hello	Bent	Bent	Bent
2	Yes/OK	Straight	Bent	Bent
3	I Need Help	Bent	Straight	Straight
4	I am Hungry	Straight	Straight	Bent
5	I am Thirsty	Bent	Bent	Straight
6	Thank You	Straight	Straight	Straight

V. CONCLUSION

This project, the Sign Language Detection Device, aims at offering a practical and user-friendly solution to bridge the gap in communication between the speech and/or hearing- impaired individuals and the general public. The system will employ flex sensors, an MPU 6050 motion sensor, Arduino Uno, and Bluetooth communication with a mobile application to effectively convert hand gestures into meaningful text and audible speech in real time. In this work, simple sensor technology and embedded systems are combined into a practical assistive device that enhances accessibility and supports inclusive communication. The system provided reliable gesture detection, smooth wireless transmission, and text-to speech output throughout the development process. This architecture is cost-effective, easy to use, and adaptive for upgrades. Even though the device can recognize only a limited set of gestures, it offers a sound basis for extension in the near future with advanced algorithms or additional sensors. Overall, this project serves its objective of developing a workable prototype that can be used by speech- impaired people and establishes how technology-based solutions can play a significant role in improving quality of life.

VI. REFERENCES

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